Derivation of the $q(\tau)$ Variational Update

Supervised Bayesian Matrix Factorization

Self-Contained Reference for `code/update_tau.R`

Andrew Walther

March 2026

Abstract

This document derives the update for the feature-specific noise precision parameters τ_j in the Supervised Bayesian Matrix Factorization model. Unlike the updates for L, F, and β — which are all Empirical Bayes Normal Means (EBNM) problems solved via the `ebnm` R package — the τ update is a *closed-form maximum likelihood estimate* (MLE).

The key mathematical ingredient is the *expected squared residual* $\bar{R}^2_{ij}$, which includes a variance correction term that accounts for posterior uncertainty in L and F. Ignoring this correction leads to systematic overestimation of τ_j (underestimation of noise). The τ update also produces the genomics ELBO proxy, used as a convergence monitor.

This is the fourth and final modular derivation document (qβ → qL → qF → qτ (this)).

Contents

1 Setup and Notation

1.1 Model and Role of τ

$$Y_{ij} = \sum_{k=1}^{K} l_{ik} f_{jk} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0,\, \tau_j^{-1}) \tag{1}$$

Each feature j has its own noise precision $\tau_j > 0$. A flat (improper) prior is placed on τ_j, so the update is a pure MLE. τ_j does *not* appear in the Cox survival term.

1.2 Notation

Symbol	Definition	R variable
$\bar{l}_{ik}$	$\mathbb{E}_q[l_{ik}]$	`EL[i,k]`
$\overline{l^2_{ik}}$	$\mathbb{E}_q[l^2_{ik}]$	`EL2[i,k]`
$\bar{f}_{jk}$	$\mathbb{E}_q[f_{jk}]$	`EF[j,k]`
$\overline{f^2_{jk}}$	$\mathbb{E}_q[f^2_{jk}]$	`EF2[j,k]`
$\hat{Y}_{ij}$	Predicted mean $\sum_k \bar{l}_{ik}\bar{f}_{jk}$	`EL %*% t(EF)`
$\bar{R}^2_{ij}$	Expected squared residual	`R2_bar[i,j]`
V_{ij}	Variance correction term	`Var_Term[i,j]`

2 ELBO Terms Involving τ_j

With a flat prior on τ_j, the ELBO contribution from feature j is:

$$\mathcal{F}_j(\tau_j) = \mathbb{E}_q[\log P(Y_{\cdot j} \mid L, F, \tau_j)] = \mathbb{E}_q\left[\sum_{i=1}^{n}\left(\frac{1}{2}\log\tau_j - \frac{\tau_j}{2}(Y_{ij} - \eta_{ij})^2\right)\right] \tag{2}$$

where $\eta_{ij} = \sum_k l_{ik} f_{jk}$. Taking the expectation:

$$\mathcal{F}_j(\tau_j) = \frac{n}{2}\log\tau_j - \frac{\tau_j}{2}\sum_i \mathbb{E}_q\left[(Y_{ij} - \eta_{ij})^2\right] \tag{3}$$

The expected squared residual $\bar{R}^2_{ij} = \mathbb{E}_q[(Y_{ij} - \eta_{ij})^2]$ is derived in Section 3. Differentiating (3) with respect to τ_j and setting to zero:

$$\frac{d\mathcal{F}_j}{d\tau_j} = \frac{n}{2\tau_j} - \frac{1}{2}\sum_i \bar{R}^2_{ij} = 0 \quad \Rightarrow \quad \boxed{\hat{\tau}_j = \frac{n}{\sum_{i=1}^{n} \bar{R}^2_{ij}}} \tag{4}$$

3 Expected Squared Residual $\bar{R}^2_{ij}$

3.1 Naive Residual (Insufficient)

If we naively plugged in posterior means:

$$(Y_{ij} - \hat{Y}_{ij})^2 = \left(Y_{ij} - \sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 \tag{5}$$

this ignores posterior uncertainty in L and F.

3.2 Full Expectation

The correct expected squared residual is:

$$\bar{R}_{ij}^2 = \mathbb{E}_q\left[\left(Y_{ij} - \sum_k l_{ik} f_{jk}\right)^2\right]$$
$$= \left(Y_{ij} - \sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 + \mathbb{E}_q\left[\left(\sum_k (l_{ik} f_{jk} - \bar{l}_{ik}\bar{f}_{jk})\right)^2\right] \tag{6}$$

The second term (the variance correction) expands via independence of rows of L and F under q:

Derivation Steps

Assuming q factorises over (i,k) pairs (mean-field):

$$\mathbb{E}_q\left[\left(\sum_k l_{ik} f_{jk}\right)^2\right] = \sum_k \mathbb{E}_q[l_{ik}^2]\,\mathbb{E}_q[f_{jk}^2] \quad \text{(cross terms vanish by independence)}$$
$$= \sum_k \overline{l_{ik}^2}\,\overline{f_{jk}^2}$$

Meanwhile:

$$\left(\mathbb{E}_q\left[\sum_k l_{ik} f_{jk}\right]\right)^2 = \left(\sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 = \sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2 + 2\sum_{k<k'} \bar{l}_{ik}\bar{f}_{jk}\bar{l}_{ik'}\bar{f}_{jk'}$$

Therefore the variance of the prediction is:

$$\mathrm{Var}_q(\eta_{ij}) = \mathbb{E}_q[\eta_{ij}^2] - (\mathbb{E}_q[\eta_{ij}])^2$$
$$= \sum_k \overline{l_{ik}^2}\,\overline{f_{jk}^2} - \left(\sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 \tag{7}$$

But this is still not the right expansion. Using the identity $\mathbb{E}[(X-c)^2] = \mathrm{Var}(X) + (\mathbb{E}[X]-c)^2$:

$$\bar{R}_{ij}^2 = (Y_{ij} - \hat{Y}_{ij})^2 + \mathrm{Var}_q(\eta_{ij})$$
$$= (Y_{ij} - \hat{Y}_{ij})^2 + \underbrace{\sum_k \overline{l_{ik}^2}\,\overline{f_{jk}^2} - \sum_k \bar{l}_{ik}^2\bar{f}_{jk}^2}_{V_{ij}} \tag{8}$$

3.3 Variance Correction Term V_{ij}

$$\boxed{V_{ij} = \sum_k \left(\overline{l_{ik}^2}\,\overline{f_{jk}^2} - \bar{l}_{ik}^2\,\bar{f}_{jk}^2\right) = (EL2\,EF2^\top)_{ij} - (EL^2\,EF^{2\top})_{ij}} \tag{9}$$

Non-negativity: Each term $\overline{l_{ik}^2}\,\overline{f_{jk}^2} - \bar{l}_{ik}^2 \bar{f}_{jk}^2 \geq 0$ because $\mathbb{E}[X^2]\mathbb{E}[Y^2] \geq (\mathbb{E}[X]\mathbb{E}[Y])^2$ by independence and Jensen's inequality. Therefore $V_{ij} \geq 0$ and $\bar{R}_{ij}^2 \geq (Y_{ij} - \hat{Y}_{ij})^2$.

Consequence: The variance correction *inflates* the expected squared residual, which *reduces* $\hat{\tau}_j$ relative to the naive estimate. Omitting V_{ij} causes systematic overestimation of precision (underestimation of noise).

Code Mapping (`update_tau.R`)

```
compute_var_term <- function(EL, EL2, EF, EF2) {
  (EL2 %*% t(EF2)) - (EL^2 %*% t(EF^2))
}
```

Matrix form explanation:

- `EL2 %*% t(EF2)`: entry (i, j) is $\sum_k \overline{l_{ik}^2 f_{jk}^2}$
- `EL^2 %*% t(EF^2)`: entry (i, j) is $\sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2$

Correction Note

Corrections R7 and R8 (from REVISED.tex):

- R7: Sign error — the variance correction term enters with a *positive* sign in $\bar{R}_{ij}^2$, not negative. The original document had an erroneous minus sign.
- R8: Subscript — the loading index should be l_{ik} (sample i, factor k), not l_{jk} which refers to a feature.

Both are fixed in the REVISED document and reflected in `code/update_tau.R`.

4 The Full Tau Update

Combining (4) and (8):

$$\bar{R}_{ij}^2 = (Y_{ij} - \hat{Y}_{ij})^2 + V_{ij} \tag{10}$$

$$\hat{\tau}_j = \frac{n}{\sum_i \bar{R}_{ij}^2} \tag{11}$$

In matrix notation:

$$R2\_bar = (Y - EL\, EF^\top)^2 + V \qquad \text{(element-wise squared + correction)} \tag{12}$$

$$\hat{\boldsymbol{\tau}} = n \,/\, \text{colSums}(R2\_bar) \qquad \text{(element-wise division)} \tag{13}$$

Key Property

Why this is NOT an EBNM problem.
The updates for L, F, and β all involve placing a prior $g(\cdot)$ on the parameter and running an EBNM solver. The τ update is different:

- τ_j has a flat (improper uniform) prior — no shrinkage.
- The optimisation is unconstrained (no prior KL term).
- The MLE has a closed form: $\hat{\tau}_j = n / \sum_i \bar{R}_{ij}^2$.
- There is no per-k loop: all K factors contribute simultaneously through $\hat{Y}_{ij} = \sum_k \bar{l}_{ik}\bar{f}_{jk}$ and V_{ij}.

Code Mapping (`update_tau.R`)

```
update_tau <- function(Y, EL, EL2, EF, EF2, tau_floor = 1e-8) {
    n        <- nrow(Y)
    Var_Term <- compute_var_term(EL, EL2, EF, EF2)          # n x p
    R2_bar   <- (Y - EL %*% t(EF))^2 + Var_Term             # n x p
    col_sums <- colSums(R2_bar)                              # p-vector
    Tau      <- n / pmax(col_sums, n * tau_floor)            # p-vector
    ...
}
```

The `pmax` floor prevents $\hat{\tau}_j = \infty$ when the residual sum is near zero (perfect reconstruction).

5 ELBO Proxy as Convergence Monitor

Substituting the MLE $\hat{\tau}_j$ back into (3):

$$\mathcal{F}_j(\hat{\tau}_j) = \frac{n}{2}\log\hat{\tau}_j - \frac{\hat{\tau}_j}{2}\sum_i \bar{R}_{ij}^2 = \frac{n}{2}\log\hat{\tau}_j - \frac{n}{2} \tag{14}$$

The genomics *ELBO proxy* (summed over all features) is:

$$\boxed{\mathcal{E}_{\text{proxy}} = \sum_j \left[\frac{n}{2}\log\hat{\tau}_j - \frac{\hat{\tau}_j}{2}\sum_i \bar{R}_{ij}^2\right]} \tag{15}$$

This is the genomics contribution $\mathbb{E}_q[\log P(Y \mid L, F, \tau)]$ evaluated at the current posterior means and the updated τ. It is monotonically increasing (or at worst non-decreasing) across CAVI iterations if the algorithm is working correctly.

Code Mapping (`update_tau.R`)

```
elbo_proxy <- sum(n / 2 * log(Tau) - Tau / 2 * col_sums)
```

6 Verification Checks

Verification Check

V1: Non-negativity. $V_{ij} \geq 0$ (element-wise), therefore $\bar{R}_{ij}^2 \geq (Y_{ij} - \hat{Y}_{ij})^2 \geq 0$. If V_{ij} contains negative entries, there is a sign error in the implementation.
V2: Perfect reconstruction. When $Y = EL\,EF^\top$ exactly and $EL2 = EL^2$, $EF2 = EF^2$: $\bar{R}_{ij}^2 = 0$, so $\hat{\tau}_j$ hits the ceiling $1/\tau_{\text{floor}}$.
V3: Correction direction. Ignoring V (using $EL2 = EL^2$, $EF2 = EF^2$) gives $V_{ij} = 0$ and hence a *larger* $\hat{\tau}_j$ (overestimates precision). The corrected estimate is always $\leq$ the biased estimate.
V4: ELBO increases. Over a sequence of CAVI iterations (L update $\to$ F update $\to$ τ update), the ELBO proxy should be non-decreasing. A decrease signals a bug in one of the update equations.
V5: V2.R consistency. The implementation should match V2.R lines 374–376 exactly:

```
Var_Term = EL2 %*% t(EF2) - EL^2 %*% t(EF^2)
Tau = n / pmax(colSums(R2_bar), n * tau_floor)
```

7 Code Mapping

Equation	Code	Type
$V_{ij} = \sum_k (\overline{l_{ik}^2 f_{jk}^2} - \bar{l}_{ik}^2 \bar{f}_{jk}^2)$	`compute_var_term()`	$n \times p$
$\bar{R}_{ij}^2 = (Y - \hat{Y})_{ij}^2 + V_{ij}$	`R2_bar`	$n \times p$
$\hat{\tau}_j = n / \sum_i \bar{R}_{ij}^2$	`n / pmax(col_sums, ...)`	p-vector
$\mathcal{E}_{\text{proxy}}$	`elbo_proxy`	scalar

8 Algorithm Summary

1. Compute variance correction (matrix operation, all K factors at once):
$$V = EL2\,EF2^\top - EL^2\,(EF^2)^\top \quad [n \times p]$$
2. Compute expected squared residuals:
$$\bar{R}^2 = (Y - EL\,EF^\top)^{\odot 2} + V \quad [n \times p]$$
3. Compute column sums and update precision:
$$\hat{\tau}_j = \max\left(\frac{n}{\sum_i \bar{R}_{ij}^2}, \frac{1}{\tau_{\text{floor}}}\right)^{-1} \qquad [p\text{-vector}]$$
4. Compute ELBO proxy for convergence monitoring:
$$\mathcal{E}_{\text{proxy}} = \sum_j \left[\frac{n}{2}\log\hat{\tau}_j - \frac{\hat{\tau}_j}{2}\sum_i \bar{R}_{ij}^2\right]$$
5. Return $\hat{\boldsymbol{\tau}}$, $\bar{R}^2$, V, $\mathcal{E}_{\text{proxy}}$.

A Notation Index

Symbol	Meaning	Dimensions
Y	Genomics data	$n \times p$
$\hat{Y}_{ij}$	$\sum_k \bar{l}_{ik} \bar{f}_{jk}$	$n \times p$
V_{ij}	Variance correction term (≥ 0)	$n \times p$
$\bar{R}^2_{ij}$	Expected squared residual	$n \times p$
$\hat{\tau}_j$	Updated noise precision	p-vector
$\mathcal{E}_{\text{proxy}}$	Genomics ELBO proxy	scalar
τ_{floor}	Precision floor (default 10^{-8})	scalar